

1D "Warm Up" - Approximation Theory

Aim: Illustrate key ideas about model accuracy in "simple" 1D context.

Q: How do interpolation and regression perform with inexact data?

$$y_i = G(x_i) + \varepsilon_i, \quad i = 0, \dots, n.$$

Interpolation. The Lebesgue constant of the point set $x_0, \dots, x_n$ bounds the maximum "amplification" of the errors!

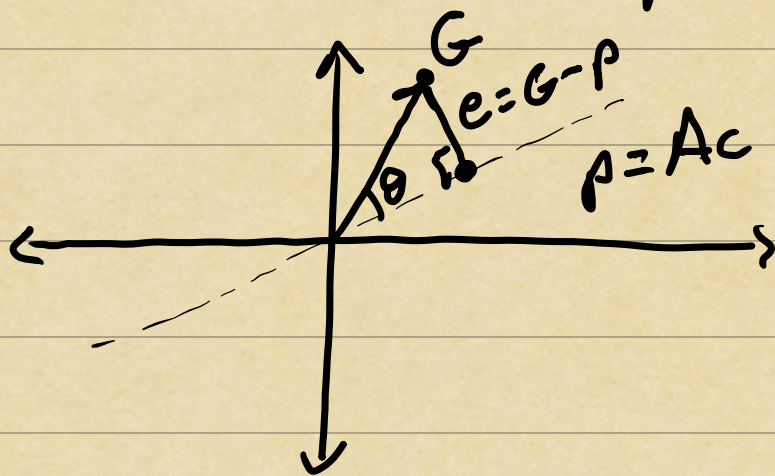
$$\|\hat{p}_n - p\| \leq \underbrace{L_n}_{\substack{\text{Lebesgue const.} \\ \hat{=} \text{error in interp.}}} \frac{\|p\|}{\|G\|} \|\hat{G} - G\| \quad \text{(relative error)}$$

$\hat{=} \text{"noise" level in data}$

Regression. If A is the "dictionary" matrix for ONB and $\sup_{0 \leq i \leq n} |\varepsilon_i|$ is small,

$$\|\hat{p} - p\| \leq \frac{\kappa(A)}{n \cos \theta} \frac{\|p\|}{\|G\|} \|\hat{G} - G\|,$$

where $\kappa = \frac{\|A\| \|c\|}{\|p\|}$ and $\theta = \cos^{-1} \frac{\|p\|}{\|G\|}$.



Ill conditioned when
 $\Rightarrow G$ nearly $\perp$ to $C(A)$.
 $\Rightarrow A$ ill-conditioned

Useful "Heuristic". (Dictionary Regression)

The sensitivity (condition #) of the computed interpolant/approximant to the perturbation $\underline{\varepsilon} = (\varepsilon_0, \dots, \varepsilon_n)^T$ is linked to the condition number

$$\kappa(A) = \frac{G_1}{G_m}, \quad A = \begin{bmatrix} e_0(x_0) & \dots & e_m(x_0) \\ \vdots & & \vdots \\ e_0(x_n) & \dots & e_m(x_n) \end{bmatrix}$$

where $G_1 \geq \dots \geq G_m > 0$ are the singular values of the dictionary matrix A .

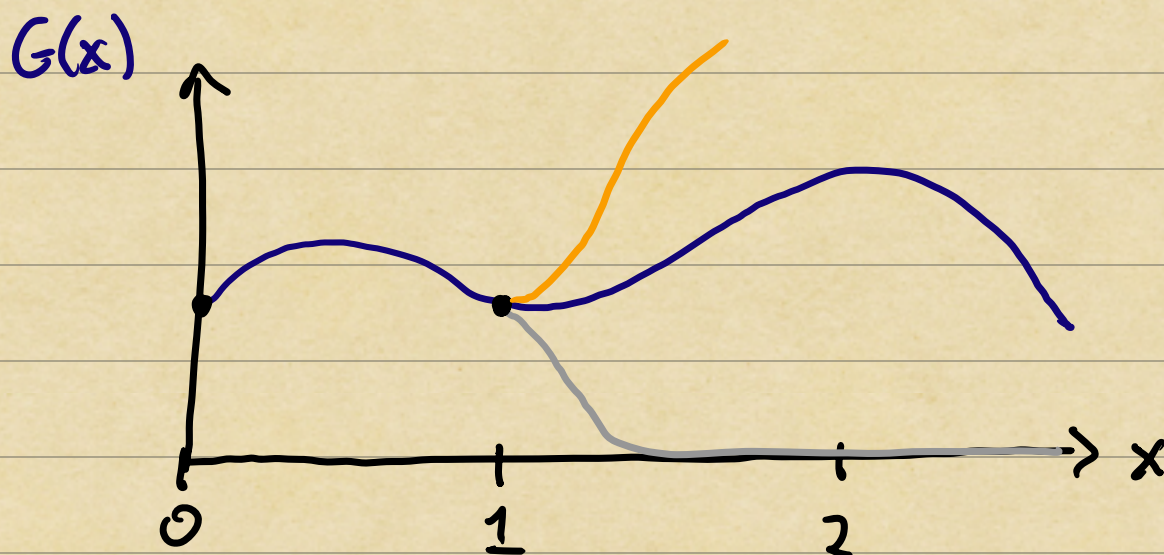
Problem 3. Extrapolation from Data

Suppose we observe a continuous function

$$y = G(x), \quad x \in \Omega \subset [0, 1],$$

where Ω could be finite or all of $[0, 1]$.

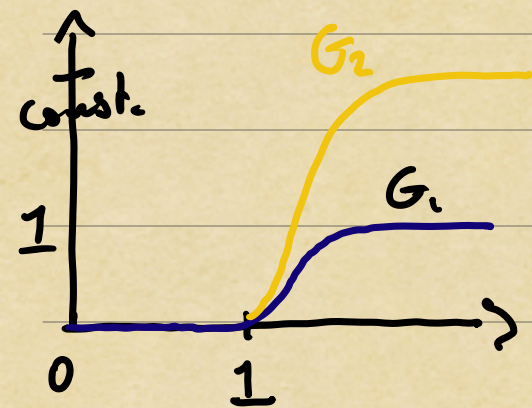
Q: What can we say about $G(x)$ when $x \in [1, b)$ for some $0 < b < \infty$?



No unique solution unless we know a lot about G . E.g., $G \in P_n$.

Q: What if we impose more smoothness?

Example: Consider the infinitely diff. functions that are 0 in $[0, 1]$:



$$G_1(x) = \begin{cases} 0, & 0 \leq x \leq 1 \\ \exp(-\frac{1}{x-1}), & 1 \leq x < \infty \end{cases}$$

$$G_2(x) = (\text{const.}) G_1(x)$$

Even $C^\infty(0, \infty)$ functions do not have a unique $C^\infty(0, \infty)$ extrapolant!

To get a unique continuation for $x \geq 1$, we need a stronger form of smoothness:

A function $G: [0, \infty) \rightarrow \mathbb{R}$ is called real analytic if for every $x_* \in [0, \infty)$, G has a power series that converges uniformly in an open interval around x_* .

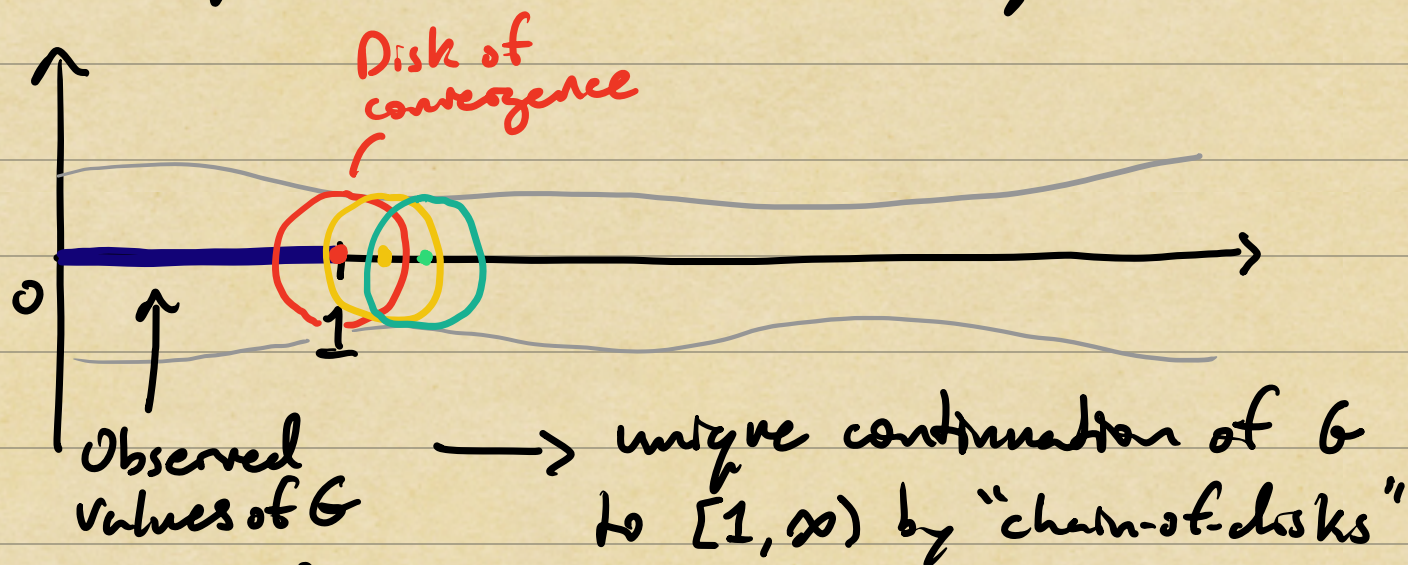
$$G(x_*) = \lim_{N \rightarrow \infty} \sum_{n=0}^N a_n (x - x_*)^n.$$

Note that $G \in C^\infty([0, \infty))$ because a uniformly convergent power series can be differentiated termwise. In particular,

$$G^{(n)}(x_0) = (n!)a_n,$$

and $|a_n|/|x-x_0|^n \rightarrow 0$ as $n \rightarrow \infty$, so the derivatives of G cannot grow "too fast".

More importantly, real analytic functions have unique analytic continuation to an open set (in $\mathbb{C}$) containing $[0, \infty)$.



Thm. If two real analytic functions $G: [0, \infty) \rightarrow \mathbb{R}$ and $F: [0, \infty)$ agree on any open subset of $\mathbb{R}$, then $G(x) = F(x)$, $\forall x \in [0, \infty)$.

$\Rightarrow$ We can construct a real analytic extension of G from $[0,1]$ to $[0,\infty)$ using the "Chain-of-disks" method:

- 1) Expand G in a power series around $x_0=1$, which converges uniformly in an open disk.
- 2) Choose a point $x_1 > 1$ in the open disk of convergence and expand G in a new power series around x_1 , using the old power series to evaluate G and its derivatives at x_1 .
- 3) Repeat the process iteratively to extend G to the whole pos. real axis $[0,\infty)$.

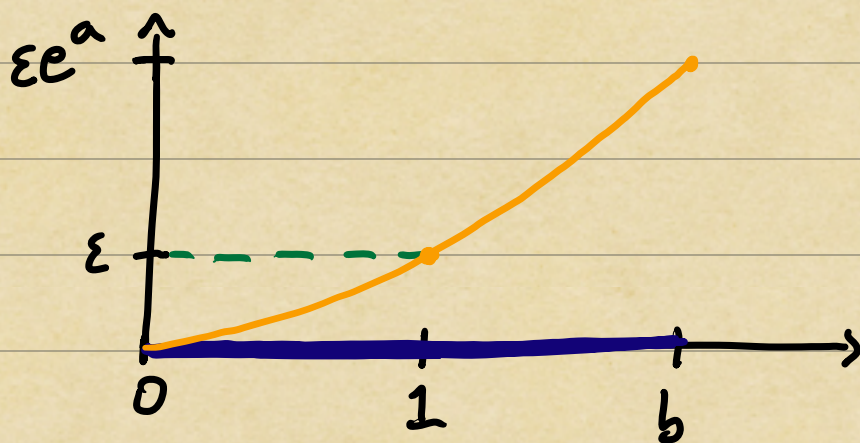
By Thm 1, the extension is unique!
However, it is **not** stable...

Example: Suppose that $f:[0, \infty)$ is real analytic and, furthermore, that there is some $\varepsilon > 0$ for which

$$|f(x)| \leq \varepsilon, \quad 0 \leq x \leq 1.$$

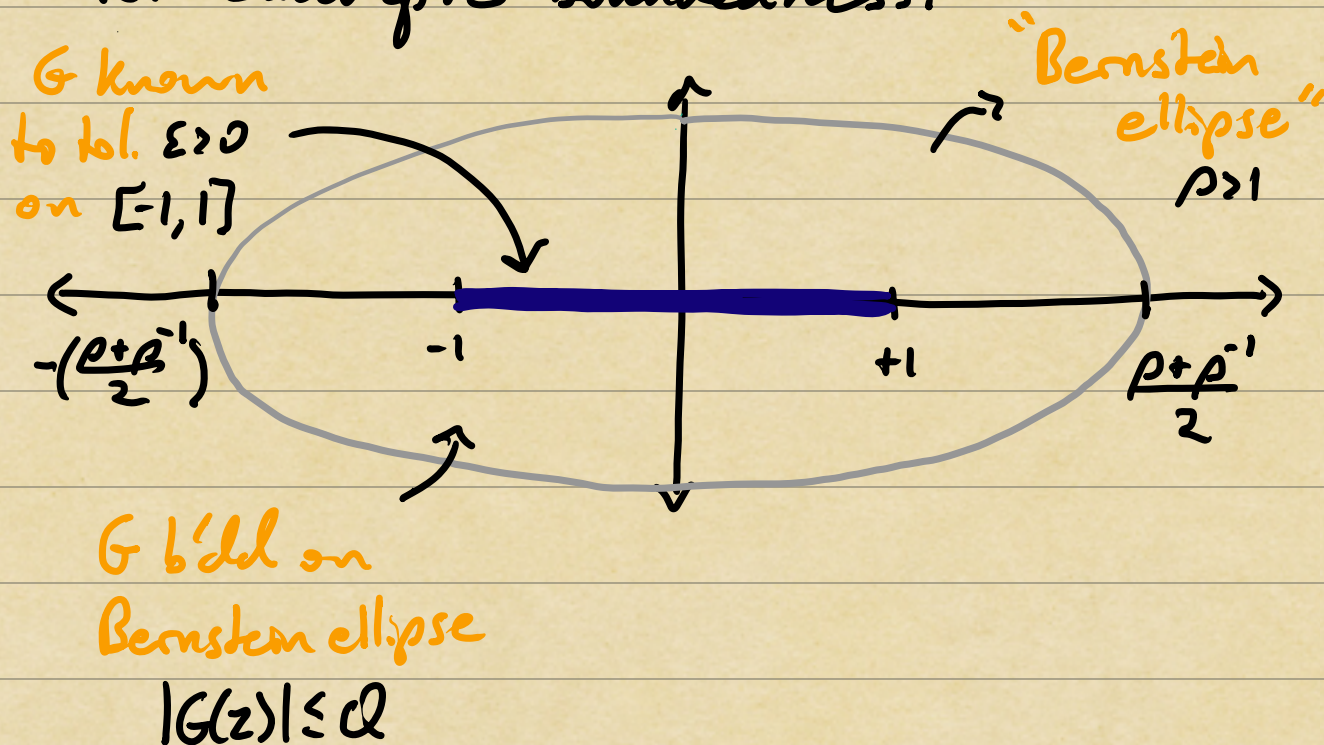
We can construct two real analytic extensions of f that differ by an arbitrary factor at any desired point $b > 1$.

$$f_1(x) = 0 \quad \text{and} \quad f_2(x) = \varepsilon \exp(-a(1 - \frac{2}{b}x)) \quad \rightarrow a \geq 0$$



By construction, $|f_2(b) - f_1(b)| = e^a$ and a can be chosen arbitrarily large.

For stable extrapolation of analytic functions, one requires additional structure in the target class of functions, for example boundedness.



$\Rightarrow$ Can compute $\tilde{G}(x)$ on $[1, \frac{\rho+\rho^{-1}}{2}]$ that satisfies

$$|G(x) - \tilde{G}(x)| \leq C_{\rho, \varepsilon} \frac{Q}{1-r(x)} \left(\frac{\varepsilon}{Q} \right)^{-\frac{\log r(x)}{\log \rho}}$$

where $r(x) = \frac{x + \sqrt{x^2 - 1}}{\rho}$, $C_{\rho, \varepsilon}$ is a const. depending polylogarithmically on ε .

Note: the bound is proportional to ε when

$x=1$ and deteriorates gradually as x increases from the edge of the interval $[-1,1]$ to the edge of the Bernstein ellipse at $x = \frac{\rho+\rho^{-1}}{2}$.

$\Rightarrow$ Can show this is essentially optimal. There are b'dd real analytic functions in the Bernstein ellipse that grow like $(1-r(x))^{-1} \varepsilon^{-\log r(x)/\log \rho}$.

$\Rightarrow$ The extrapolant can be computed from equispaced data on $[-1,1]$ provided that an oversampling criterion is met.

For more details, see the reading attached to Lecture 7,

"Stable Extrapolation of Analytic Functions"
by Laurent Demchenko and Alex Townsend

"Quantifying the Ill-Conditioning of Analytic Continuation" by Nick Trefethen